

Verminator Verilator-Based Fault-Injection Framework

HM | ISOLDE | T4.3 / D4.5 Section 3.4.1 | Version ISOLDE-D4.5-2026.07

Schema Version			0.9
Basic Information			
IP Name		Verminator Verilator-Based Fault-Injection Framework	
Version		ISOLDE-D4.5-2026.07	
Status		Prototype implemented	
Comment		Draft metadata release derived from ISOLDE D4.5. The value in basicInfo.version identifies this card release, not necessarily the partner's product or repository tag. Partner confirmation is required before publication.	
Provider	Item 1 / Organization Name	Munich University of Applied Sciences	
Provider	Item 1 / Organization URL	https://www.hm.edu/	
Provider	Item 1 / Organization Alias	• HM	
License		Other	
Repository URL		https://github.com/hm-aemy/verminator	
Project		ISOLDE	
Work Item		T4.3 / D4.5 Section 3.4.1	
IP Type	Category	• Software Tool • Fault Simulation • V&V Tools	
IP Type	Domain	• Safety	
IP Type	Comment	Verilator AST instrumentation and configurable fault-injection campaigns	
IP Type	Service	no	
License Details		Mixed open-source licensing reported as LGPL-3.0 and Apache-2.0. File-level licensing applies.	
System-Level Information			
Architecture		Generic RTL with RISC-V demonstrator use	
Processor		Design-dependent	
ISAExtension		Not applicable	
Board		Host-based simulation	
SoC		SystemVerilog/Verilog design under test	
SW Dependencies		Modified Verilator, C++ toolchain, SystemVerilog DPI and campaign configuration	
Requires Specialized Tool		Yes. The Verminator Verilator extension and compatible C++ build flow.	
Interfaces and Artifacts			

CLI		• Build the instrumented model with the Verminator-enabled Verilator • Execute a configured fault-injection campaign
API		• SystemVerilog DPI fault-control hooks • Custom C++ campaign-control integration
Input Artifacts	Item 1 / Artifact	RTL design
Input Artifacts	Item 1 / Format	SystemVerilog/Verilog
Input Artifacts	Item 2 / Artifact	Fault-injection configuration
Input Artifacts	Item 2 / Format	Configuration files
Input Artifacts	Item 3 / Artifact	Optional custom campaign controller
Input Artifacts	Item 3 / Format	C++ source
Output Artifacts	Item 1 / Artifact	Instrumented simulation model
Output Artifacts	Item 1 / Format	C++/executable
Output Artifacts	Item 2 / Artifact	Fault analysis report
Output Artifacts	Item 2 / Format	Report/log
Output Artifacts	Item 3 / Artifact	Simulation results
Output Artifacts	Item 3 / Format	Logs and result data
Deployment		
Bundle		Open-source Git repository containing the Verilator extension and campaign framework
Project-Specific Features		
Purpose		Add simulation-based fault injection to Verilator through AST manipulation and DPI hooks.
Validated Capabilities		• DPI-hook insertion • Custom fault-model support • Executable simulation generation • Manual user-defined campaign execution
Limitations		• Literal and implicit HDL types/signals are limited to a maximum width of 64 bits. • Some automated analysis and preset features were incomplete at the D4.5 cut-off.
Public Repository Status		Public
Source Basis	Deliverable	ISOLDE D4.5 - Software for Hardware Design and Validation Implementation
Source Basis	Section	3.4.1 Verilator-based Fault-Injection Framework
Source Basis	SW IP Card Model	Schema v0.9 with basicInfo, systemLevelInfo, interfaces, deployment and features
Requirements Summary	Count	8

Requirements Summary	Ids	• FI-RTL-01 • FI-RTL-02 • FI-RTL-03 • FI-RTL-04 • FI-RTL-01 • FI-RTL-02 • FI-RTL-03 • FI-RTL-04
Metadata Review	Partner Review Required	true
Metadata Review	Items	• Replace the metadata release identifier with the exact product/repository version where available. • Confirm public repository or product URL. • Confirm licensing and redistribution wording. • Add a tool-specific TRL only if the partner has approved it.